

An **RC circuit** consists of a **resistor** R and a **capacitor** C connected in series with a voltage source (electromotive force) $\mathcal{E}$. The capacitor stores energy in its electric field and **resists sudden changes in voltage** across its plates.

Basic relations

The basic voltage–current and capacitor relations are

$$V_R = IR, \quad V_C = \frac{Q}{C}, \quad I = \frac{dQ}{dt},$$

where Q is the capacitor charge, V_C the capacitor voltage, and I the circuit current. Because $Q = CV_C$, the current may also be written as

$$I = C \frac{dV_C}{dt}.$$

KVL and governing differential equation

Applying Kirchhoff's Voltage Law (KVL) around the loop gives

$$\mathcal{E} - V_R - V_C = 0.$$

Substituting $V_R = IR$ and $I = C \frac{dV_C}{dt}$ yields

$$\mathcal{E} - R \left(C \frac{dV_C}{dt} \right) - V_C = 0.$$

Rearrange to obtain the first-order linear ODE for the capacitor voltage:

$$\boxed{RC \frac{dV_C}{dt} + V_C = \mathcal{E}}.$$

Solution: Charging (initially uncharged capacitor)

Assume the capacitor is uncharged at $t = 0$, i.e. $V_C(0) = 0$. The ODE

$$\frac{dV_C}{dt} + \frac{1}{RC} V_C = \frac{\mathcal{E}}{RC}$$

has the solution (standard integrating factor / linear ODE):

$$\boxed{V_C(t) = \mathcal{E} \left(1 - e^{-t/(RC)} \right)}.$$

The circuit current follows from $I = C \frac{dV_C}{dt}$:

$$\boxed{I(t) = C \frac{dV_C}{dt} = \frac{\mathcal{E}}{R} e^{-t/(RC)}}.$$

Remarks

- $V_C(\infty) = \mathcal{E}$ (steady-state). The capacitor behaves like an open circuit to steady DC.
- $I(0) = \frac{\mathcal{E}}{R}$ (initial current). At the instant the switch is closed the capacitor acts like a short circuit.
- The **time constant** is

$$\boxed{\tau = RC}$$

and after $t = \tau$ the capacitor voltage reaches $1 - e^{-1} \approx 0.632$ (about 63.2%) of its final value.

Discharging (no source)

If a capacitor charged to voltage V_0 at $t = 0$ is disconnected from the source and allowed to discharge through R , KVL gives

$$RC \frac{dV_C}{dt} + V_C = 0.$$

The solution is

$$V_C(t) = V_0 e^{-t/(RC)}, \quad I(t) = -\frac{V_0}{R} e^{-t/(RC)}$$

(the negative sign for I indicates the current flows opposite the charging direction).

Energy stored

The energy stored in the capacitor's electric field is

$$E_{\text{cap}} = \frac{1}{2} C V_C^2.$$

During charging substitute $V_C(t) = \mathcal{E}(1 - e^{-t/RC})$ to get the instantaneous stored energy. During discharging this energy is dissipated as heat in the resistor R .

Key observations (summary)

1. **Initial behavior (charging):** $V_C(0) = 0$, $I(0) = \mathcal{E}/R$. The capacitor initially behaves like a short circuit.
2. **Steady-state behavior:** $V_C(\infty) = \mathcal{E}$, $I(\infty) = 0$. The capacitor behaves like an open circuit to DC.
3. **Time constant:** $\tau = RC$. Larger R or C slows charging/discharging.
4. **Complement to RL circuits:** An inductor resists changes in current; a capacitor resists changes in voltage.
5. **Applications:** Timing circuits, filters (low-pass/high-pass), coupling/decoupling, pulse shaping.

Relations:

$$V_R = IR, \quad V_C = \frac{Q}{C}, \quad I = \frac{dQ}{dt} = C \frac{dV_C}{dt}.$$

KVL:

$$\mathcal{E} - IR - V_C = 0 \quad \Rightarrow \quad RC \frac{dV_C}{dt} + V_C = \mathcal{E}.$$

Charging ($V_C(0) = 0$):

$$V_C(t) = \mathcal{E}(1 - e^{-t/(RC)}), \quad I(t) = \frac{\mathcal{E}}{R}e^{-t/(RC)}, \quad Q(t) = C\mathcal{E}(1 - e^{-t/(RC)}).$$

$$\tau_{RC} = RC.$$

Discharging ($V(0) = V_0$):

$$V_C(t) = V_0 e^{-t/(RC)}, \quad I(t) = -\frac{V_0}{R}e^{-t/(RC)}.$$

Energy:

$$U = \frac{1}{2}CV^2 = \frac{Q^2}{2C}.$$

Notes: Initial current at switch-on is $\mathcal{E}/R$. The capacitor behaves like a short at $t = 0$ and as an open circuit at steady DC. Use RC to estimate transient durations.